

Problem 1

Given:

$$T_{clk} = 4ns$$

$$T_{cq} = 0.5ns$$

$$T_{setup} = 0.5ns$$

Path 1: Input to FF1

$$Arrival = Input\ delay + Data\ delay$$

$$Arrival = 0.5 + 2 = 2.5ns$$

$$Required = T_{clk} - T_{setup}$$

$$Required = 4 - 0.5 = 3.5ns$$

$$Slack = Required - Arrival$$

$$Slack = 3.5 - 2.5 = 1ns$$

Path 2: FF1 to FF2

$$Arrival = T_{cq} + Combinational\ delay$$

$$Arrival = 0.5 + 2 = 2.5ns$$

$$Required = 4 - 0.5 = 3.5ns$$

$$Slack = 3.5 - 2.5 = 1ns$$

Path 3: FF2 to Output

$$Arrival = 0.5 + 2.4 = 2.9ns$$

$$Required = 4 - 0.5 = 3.5ns$$

$$Slack = 3.5 - 2.9 = 0.6ns$$

Path 4: Input to Output

$$Arrival = 0.5 + 1 = 1.5ns$$

$$Required = 4 - 0.5 = 3.5ns$$

$$Slack = 3.5 - 1.5 = 2ns$$

Worst Timing Slack

$Worst\ Slack = +0.6ns$

Problem 2

Given:

$$T_{clk} = 10ns$$

Clock Skew Calculation

$$Clock\ Skew = (3 + 9 + 3) - 2$$

$$Clock\ Skew = 13ns$$

Maximum Data Delay

$$T_{data(max)} = T_{cq(max)} + Data\ path$$

$$T_{data(max)} = 11 + (2 + 9 + 2)$$

$$T_{data(max)} = 24ns$$

Setup Check

$$Required = T_{clk} + Skew - T_{setup}$$

$$Required = 10 + 13 - 4$$

$$Required = 19ns$$

$$Slack = Required - Arrival$$

$$Slack = 19 - 24 = -5ns$$

$$\boxed{Setup\ Violation = 5ns}$$

Hold Check

$$T_{data(min)} = 9 + (1 + 6 + 1)$$

$$T_{data(min)} = 17ns$$

$$Required = Clock\ Skew + T_{hold}$$

$$Required = 13 + 2 = 15ns$$

$$Slack = 17 - 15 = 2ns$$

$$\boxed{Hold\ Slack = +2ns}$$

Final Result

$$\boxed{Only\ Setup\ Violation\ Exists}$$